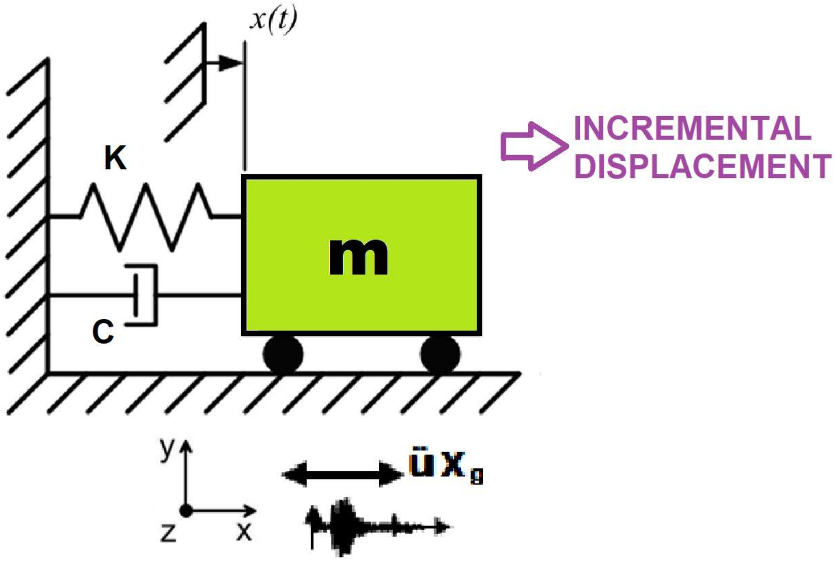
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

DETERMINATION OF OPTIMUM STRUCTURAL INITIAL ELASTIC STIFFNESS FROM STRUCTURAL PERIOD USING FINITE-DIFFERENCE NEWTON ITERATION AND OPENSEES

THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

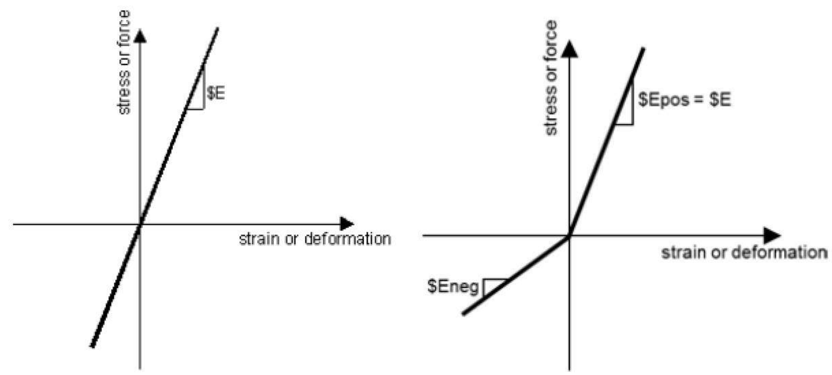
$$\Delta_d = \text{Lateral Displaement from Dynamic Analysis}$$

$$\Delta_y = \text{Lateral Yield Displaement from Pushover Analysis}$$

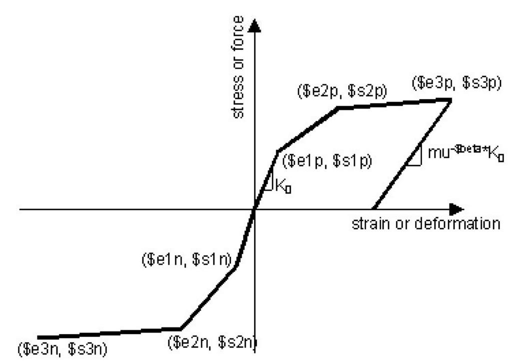
$$\Delta_u = \text{Lateral Ultimate Displaement from Pushover Analysis}$$

$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$

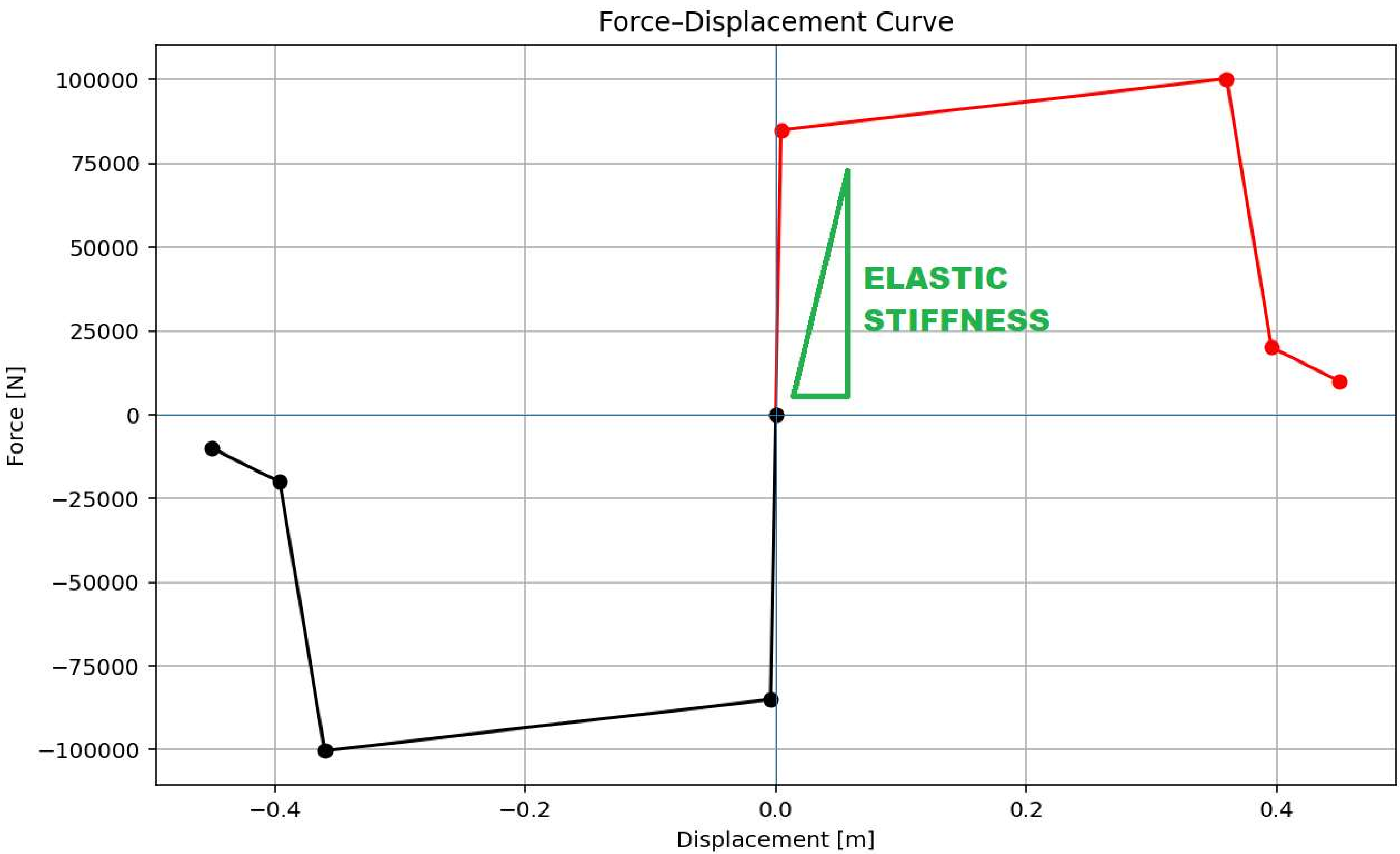
$$m\ddot{u} + c\dot{u} + ku = P(t)$$



ELASTIC MATERIAL

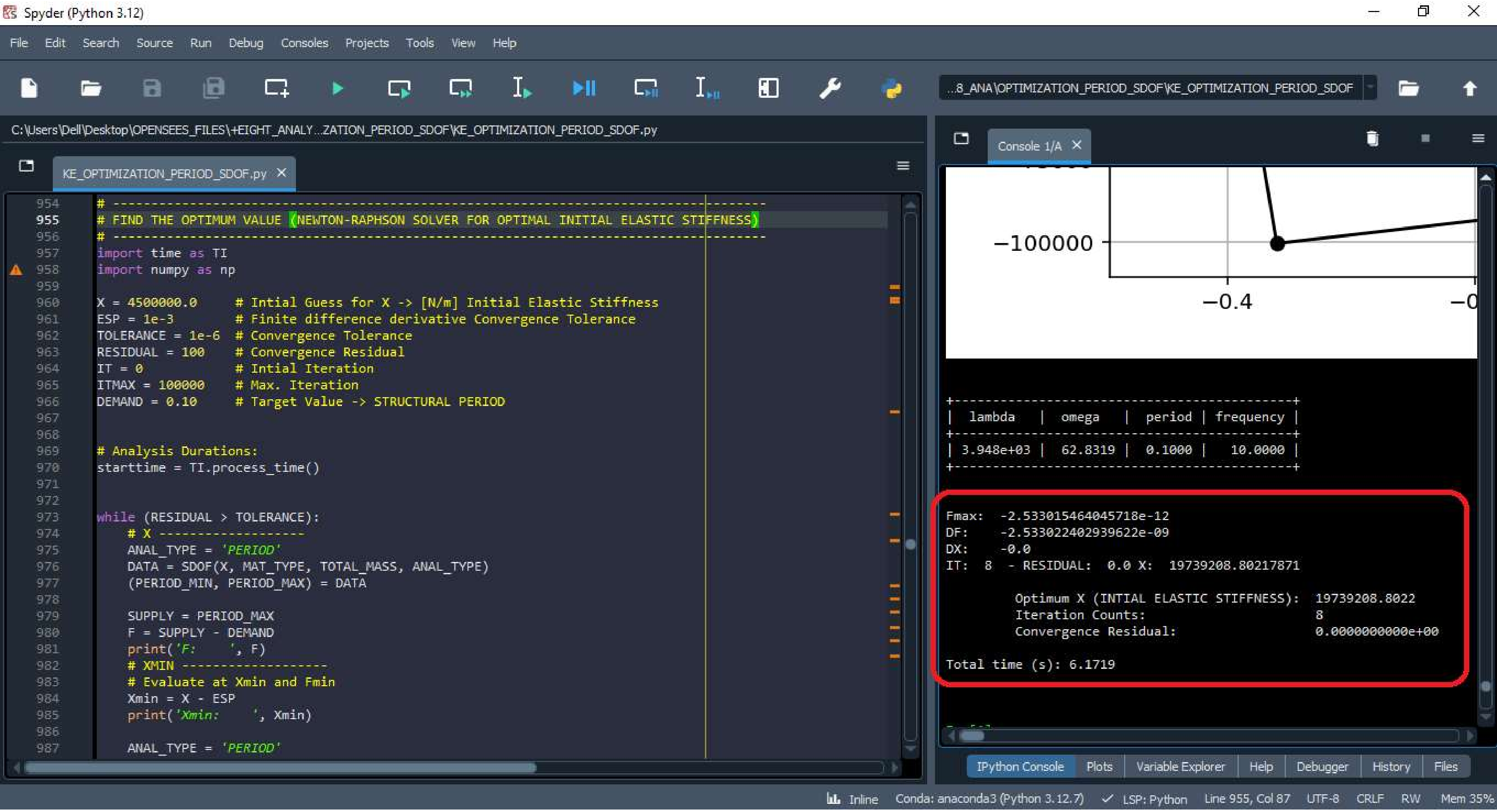


INELASTIC MATERIAL



Force–Displacement Curve
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
Force [N]
−0.4
−0.2
0.0
0.2
0.4
Displacement [m]
ELASTIC
STIFFNESS

OPTIMIZATION PROBLEM FOR STRUCTURAL PERIOD BY CHANGING INTIAL ELASTIC STIFFNESS


Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...ZATION_PERIOD_SDOF\KE_OPTIMIZATION_PERIOD_SDOF.py
KE_OPTIMIZATION_PERIOD_SDOF.py
# --
# FIND THE OPTIMUM VALUE (NEWTON-RAPHSON SOLVER FOR OPTIMAL INITIAL ELASTIC STIFFNESS)
# --
import time as TI
import numpy as np
X = 4500000.0   # Intial Guess for X -> [N/m] Initial Elastic Stiffness
ESP = 1e-3   # Finite difference derivative Convergence Tolerance
TOLERANCE = 1e-6 # Convergence Tolerance
RESIDUAL = 100   # Convergence Residual
IT = 0   # Intial Iteration
ITMAX = 100000   # Max. Iteration
DEMAND = 0.10   # Target Value -> STRUCTURAL PERIOD
# Analysis Durations:
starttime = TI.process_time()
while (RESIDUAL > TOLERANCE):
    # X ------------------
    ANAL_TYPE = 'PERIOD'
    DATA = SDOF(X, MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
    (PERIOD_MIN, PERIOD_MAX) = DATA
    SUPPLY = PERIOD_MAX
    F = SUPPLY - DEMAND
    print('F: ', F)
    # XMIN ------------------
    # Evaluate at Xmin and Fmin
    Xmin = X - ESP
    print('Xmin: ', Xmin)
    ANAL_TYPE = 'PERIOD'
Console 1/A
−100000
−0.4
| lambda | omega | period | frequency |
| 3.948e+03 | 62.8319 | 0.1000 | 10.0000 |
Fmax: -2.533015464045718e-12
DF: -2.533022402939622e-09
DX: -0.0
IT: 8 - RESIDUAL: 0.0 X: 19739208.80217871
Optimum X (INTIAL ELASTIC STIFFNESS): 19739208.8022
Iteration Counts: 8
Convergence Residual: 0.0000000000e+00
Total time (s): 6.1719
IPython Console Plots Variable Explorer Help Debugger History Files
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 955, Col 87 UTF-8 CRLF RW Mem 35%

```
+------------------------------------------+
| lambda  | omega  | period | frequency |
+------------------------------------------+
| 3.948e+03 | 62.8319 | 0.1000 | 10.0000 |
+------------------------------------------+
```

```
Fmax:  -2.533015464045718e-12
DF:    -2.533022402939622e-09
DX:    -0.0
IT:  8  - RESIDUAL:  0.0 X:  19739208.80217871
```

```
        Optimum X (INTIAL ELASTIC STIFFNESS):  19739208.8022
        Iteration Counts:                  8
        Convergence Residual:              0.0000000000e+00
```

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Total time (s): 6.1719
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